

CHEMISTRY ASSIGNMENT CLASS SEMINAR

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B Tech CSE

C32

PARTICULATE POLLUTANTS

Q. What is pollutants?

=> A substance that **pollutes** something like **air, water, soil**.

Q. What does pollute mean?

=> **Contaminating** – with **harmful** substance. Excessive amount of **metal** and **metal oxides** in nature.

Air borne small solid particles and liquid droplets are commonly known as particulates.

Basically, **any minute particle** is known as particulate.

They vary in size ranging from a **diameter of 0.0002 μ to 500 μ** . Particulates possess large surface areas and hence appear to form attractive sites for sorption of various organic and inorganic matters.

Types of particulates

Particulate pollutants may be classified according to their **optical properties, nature** and **size** as follows:

(i) Smoke

Smoke particulates **consist of solid and liquid particles** ranging from **0.05 μ to 1.0 μ** which are formed during **incomplete combustion** of carbonaceous materials.

e.g. It includes smoke of gaseous pollutants like oxides of Sulphur (50.00 $\mu\text{g}/\text{m}^3$), oxides of nitrogen, carbon monoxide and hydrocarbons.

(ii) Dust

Dust is composed of fine **solid particles** and their size range from **1 μ to 100 μ** . These particles are formed by

(a) entrained materials in mechanical processing like sand from sand blasting (b) the particles directly from materials when it undergoes a mechanical operation like saw dust from wood works and

(c) directly from the material being processed like coal.

(iii) Mist

Mist or liquid particles are formed by **condensation of vapor**, having a size less than **10 μ** . 22 °C

e.g. SO₂ (g) SO₂ (l)

(iv) Spray

It constitutes **liquid particles** obtained from the parent liquid by the process of mechanical disintegration like atomization.

Q. How can water droplets / spray can be a pollutant?

=> Water droplets are carriers of micro-organisms and viruses.

(v) Fumes

Fumes are generally obtained by the **condensation of vapors** by **sublimation, distillation, boiling, calcination** and by several other chemical reactions.

e.g. Generally organic solvents, metals and metallic oxides form fume particles having less than **1 μ** .

Sources of Particulates

(a) Natural sources

- **volcanic eruption**- Ninety-nine percent of the gas molecules emitted during a volcanic eruption are water vapor (H₂O), carbon dioxide (CO₂), and sulfur dioxide (SO₂)
- blowing of dust and soil by the wind
- spraying of salt and various other solid particles by the seas and oceans
- natural process injects nearly 800-2000 million tones of particulate matter each year into atmosphere

(b) Man made sources

- combustion of fossil fuels
- industrial operation like mining and polishing
- industries related with the production of pesticides, fertilizers, textiles and chemicals
- industrial fugitive processes
- non industrial fugitive processes such as construction of buildings bridges, dams
- agricultural activities, roadways dust, fire dust
- transportation sources
- fly ash from thermal power plants
- man made activities emit nearly **450 million tons** of particulates every year into the atmosphere

According to the composition, particulate is divided into two types:

(i) **Organic** particulate (ii) **Inorganic** particulate

(i) **Organic Particulate matter** (OPM)

It includes polycyclic aromatic hydrocarbons (PAH) like chrysene, benzo fluoranthene, benzo α -pyrene, benzidine etc. The size of OPM is **20 $\mu\text{g}/\text{m}^3$** .

Sources: OPM originates mainly from **combustion of fuels, automobiles and vegetation**.
(decomposition – production of CO₂)

Effect: (i) OPM is mainly **carcinogenic** in nature.

(ii) PAH compounds remain absorbed in soot particles. Soot particles act as a disease carrier containing toxic trace metals like Be, Cr, Mn, Ni, U, Cd, Fe and V.

Polycyclic aromatic hydrocarbons (PAHs) are a class of chemicals that occur naturally in coal, crude oil, and gasoline. They result from burning coal, oil, gas, wood, garbage, and tobacco.

(ii) **Inorganic particulate matter**

Inorganic particulate includes **aerosols** and **fly ash**.

(a) **Aerosol:**

Smaller solid particles in the liquid droplets are called as aerosol.

Sources:

- Sea spray and dust are natural sources of aerosol.
- Man made activities of aerosols are from Industrial combustion and condensed organic vapor.
- The size of aerosols ranges from 1 nm to 100 µm in diameter.
- Aerosols include dust, smoke, fly ash, pollen grains, cloud, mist, fog, smog.

Condensed aerosol:

The gases which are present in the atmosphere undergo chemical reactions and forms liquid of solid particles and it is known as condensed aerosol. **Cloud, mist, smog** are the examples for condensed aerosol.

Sea Sprays:

Waves are produced in the sea due to the vigorous action of wind and number of bubbles are formed and collapsed frequently. This is known as **white cap** or **sea spray** or **sea salt aerosol**.

Aerosol dust:

Solid materials like clay minerals and finely divided organic matter undergo fragmentation by the action of wind and mixed with atmosphere and are known as aerosol dust.

Aerosol mist:

They are formed by the **oxidation of SO₂ to H₂SO₄ by dust** and forms cloud.



Tetraethyl lead which is present in leaded gasoline combines with oxygen and alkyl halides and forms condensed aerosols.

Effects of Aerosol

- It produces respiratory problems
- It reduces the visibility
- It is responsible for acid rain, photochemical smog

(b) Fly ash

The smaller ash particle which enters into the atmosphere and pollutes the air is known as fly ash.

Source

- combustion of fossil fuel
- industrial combustion processes
- ash contains the oxides of inorganic particulate matter

Effects

- it is carcinogenic in nature
- they cause asthma and respiratory problems